Real-world Fern Identification

From research benchmark to field-ready AI — a falsification framework

FIELD PHOTOGRAPH



Natural scene · uncontrolled framing

Photo: Cynthia Su · CC BY

HERBARIUM SHEET



Standardized specimen · machine-readable label

US National Herbarium, Smithsonian Institution · CC0

Cyrtomium macrophyllum (Dryopteridaceae)

CONVENTIONAL BENCHMARK

91.4 ± 0.7%

5,969 images · 152 species · 30 families

Can we trust this number?

1 Both regimes represented during training



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BOTH REGIMES IN TRAINING

HERBARIUM	FIELD
0.9198	0.9250

91.98% · 92.50%

2 Domains separated between training and test

TRAIN FIELD ✖ TEST HERBARIUM

TRAIN HERBARIUM ✖ TEST FIELD

0.4238–0.6538

42.38%–65.38%

when the two regimes are separated between training and test

The performance collapse is real

	MIXED-REGIME TRAINING	SEPARATED TRAINING → TEST
HERBARIUM	0.9198	0.4238–0.6538
FIELD	0.9250	0.4238–0.6538

3 Three hidden assumptions

Image domain assumption
Training and test images share the same distribution.

Geography assumption
Species, locations and environments are consistent across splits.

Text leakage assumption
Metadata, labels or context may leak across splits.

4 Three controlled falsification tests

T1 Image domain shift
Mixed vs. separated training/test.

T2 Geography shift
Cross-region generalization.

T3 Text leakage
Remove metadata and contextual cues.

5 Benchmark ≠ Deployment

High benchmark accuracy
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Field-ready identification

6 Open-set rejection and selective acceptance

ACCEPT — High confidence in-domain → ABSTAIN — Uncertain / low confidence → REJECT — Out-of-domain / novel species → HUMAN REVIEW — Expert verification for critical cases

7 Deployment decision pipeline

Field data → Model inference → Risk assessment → Final decision (accept / abstain / reject / review)